

#19-R: GC-GISMo — Health-Aware, Controllable Ingredient Substitution

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Problem Statement

- Task**
 - Given source s + recipe r , predict a substitute preserving flavor and satisfying user goal $g = [\text{low_sugar?}, \text{low_sodium?}]$.
- Why a new method?**
 - Existing models (FlavorGraph [1], GISMo [2]) optimize flavor/context but cannot condition on a health goal.
 - Post-hoc filtering on GISMo top-K hurts retrieval and can't respond to user g .
- Our contribution**
 - GC-GISMo: nutrient-hub graph + goal vector g + hinge L_{health} . One model $\rightarrow$ low-sugar / low-sodium / both.
- References**
 - [1] Park, Donghyeon, et al. "FlavorGraph: a large-scale food-chemical graph for generating food representations and recommending food pairings." Scientific reports 11.1 (2021): 931.

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Limitations of Existing Methods

- Existing methods preserve flavor or context — but not health goals**
 - FlavorGraph [1]: Similarity only, no substitution mechanism.
 - GISMo: Context-aware substitution, but health-unaware.
 - Post-hoc filter: Trades MRR for health, ignores g .

Capability	FlavorGraph	GISMo	GC-GISMo (ours)
Ingredient substitution	No (similarity only)	Yes	Yes
Flavor-preserving	Yes (I-F edges)	Yes (FlavorGraph)	Yes (FlavorGraph)
Recipe context-aware	No	Yes (mean c_r)	Yes (mean c_r)
Nutrition-aware graph	No	No	Yes
Health-goal conditioned	No	No	Yes (goal vector)
One model, multi-target	N/A	N/A	Yes (override g)

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Our Approach: GC-GISMo

- Pipeline**

```
graph LR
    R[Recipe r] --> GE[GIN Encoder  
2-layer GINConv  
with edge-weighted  
aggregation  
embed_dim = 300]
    S[Source s] --> GE
    GE --> Hs[h_s  
Source embed]
    GE --> Hv[h_v  
Candidate embed]
    GE --> Cr[c_r  
Mean of recipe]
    Hs --> DMLP[Decoder MLP  
3-layer MLP on  
[h_s || h_v || c_r || g]]
    Hv --> DMLP
    Cr --> DMLP
    G[Goal g] -.-> TL[Training Loss L_total]
    DMLP -.-> TL
```

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Dataset & Graph Architecture (GC-GISMo)

- Dataset**
 - FlavorGraph: 6,313 ingredients + 1,645 flavor compounds + NPMI (Normalized Pointwise Mutual Information) edges
 - Recipe1MSubs: 48,229 / 10,552 / 10,548 train / val / test pairs
 - USDA FoodData Central: 7-nutrient per-ingredient mapping (ours)
- Heterogeneous Graph (GC-GISMo Full)**
 - Nodes**
 - 6,313 ingredients (I) + 1,645 flavor compounds (F) + 7 nutrient hubs (N) (sugar, sodium, fat, saturated-fat, carbohydrate, protein, calories).
 - Edges propagate through WeightedGINConv:**
 - I-I: NPMI co-occurrence (Recipe1MSubs)
 - I-F: FlavorDB compound bindings
 - I-N (ours): $\log 1p$ + per-nutrient min-max

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Loss Function & Training

- $\mathcal{L} = \mathcal{L}_{\text{sub}} + \lambda_h \cdot \mathcal{L}_{\text{health}}$
- $\mathcal{L}_{\text{sub}}$: **GISMo InfoNCE**
 - $\mathcal{L}_{\text{sub}} = -\log \left(\frac{e^{\phi(s, y, r, g)}}{\sum_{c \in C} e^{\phi(s, c, r, g)}} \right)$
 - Candidate set $C = \{ \text{target } y, K=10 \text{ random negatives} \}$.
- $\mathcal{L}_{\text{health}}$: **hinge on expected nutrient gap**
 - $\mathcal{L}_{\text{health}} = \sum_k g_k \cdot \max(0, m - \mathbb{E}_{v \sim p_\theta} [(n_s - n_v)_k])$
 - Softmax expectation $\rightarrow \nabla \phi(s, v, r, g)$.
- $\lambda_h = 1.0$ is the sweet spot

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Evaluation

- Models**
 - Baseline / Baseline with Health Filter (GISMo_HF)
 - GC-GISMo: structural nutrient-hub injection — main model
 - GC-GISMo (encoder inj.) / (decoder concat)
 - Goal injected at encoder / decoder only
- Metrics**
 - MRR**: Mean Reciprocal Rank of the ground-truth substitute.
 - Hit@10**: Ground-truth in top-10 rate.
 - sugar/sodium_sat**: Rate of recommendations with lower (or equal) sugar_g (sodium_mg) than source.
 - ID_MRR**: MRR on (source, target) pairs seen during training.
 - OOD_MRR**: MRR on unseen pairs; measures generalization.

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Results

- GC-GISMo: best retrieval across all methods**
- (MRR +6.8, Hit@10 +9.1 vs. Vanilla GISMo)

Method	MRR	Hit@10	sugar_sat	sodium_sat	ID_MRR	OOD_MRR
Baselines						
Vanilla GISMo	26.46	48.69	46.98	43.91	37.74	5.71
GISMo_HF	27.83	50.50	50.31*	50.27*	39.52	6.31
Ours						
GC-GISMo	33.25	57.77	46.00	44.84	46.42	9.02
GC-GISMo (encoder inj.)	32.34	54.82	47.56	45.89	45.53	8.04
GC-GISMo (decoder concat)	31.29	53.78	45.85	45.03	43.74	8.34

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Conclusion

- Qualitative comparison (Source=flour, GT=breadcrumb)**

Goal g	Substitution	d_sugar(g)	d_sodium(mg)
auto (data-derived)	wheat_flour	+0.00	+0
$g = [1, 0]$ low-sugar	cornstarch	+1.66	-5
$g = [0, 1]$ low-sodium	tapioca_starch	-85.01	+4
$g = [1, 1]$ both	whole_wheat_flour	+0.64	+1

- Conclusion**
 - GC-GISMo trades some **health-sat** for **retrieval gains**.
 - Health-sat** can be recovered via **goal vector g** .

Goal g	sugar_sat	sodium_sat	MRR
auto (data-derived)	46.00	44.84	33.25
$g = [1, 0]$ low-sugar	73.25	6.19	13.02
$g = [0, 1]$ low-sodium	13.57	80.96	12.36
$g = [1, 1]$ both	71.65	81.77	13.72

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